

Multi-Agent Scientific Paper Review

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File	Final_paper_kjns_210426 (1).pdf
Words	12,275
Sections Detected	preamble, introduction, methodology, experiments, discussion, section_vi, conclusions, section_vii, biogra
Review Mode	DEEP
Model	qwen3:8b

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Editorial Decision

MINOR REVISION

Weighted Score: 3.8 / 5.0
Confidence: 87%

Executive Summary

The manuscript presents a novel physics-informed EANN framework for structural parameter identification, with well-documented methodology and robust statistical validation. However, it requires significant revisions to address critical issues in structure, literature review, and ethical reporting.

Decision Rationale

The manuscript has significant strengths in methodology, statistical validation, and experimental setup, but it requires major revisions to address critical issues in structure, literature review, and ethical reporting. These revisions are necessary for publication in a Q1 journal.

Evaluation Score Table

Scale: 0 = Not evaluable | 1 = Very weak | 2 = Weak | 3 = Acceptable | 4 = Good | 5 = Excellent

Criterion	Score	Rating	Visual
Originality & Novelty	5.0	Excellent	■■■■■
Manuscript Structure	2.5	Weak	■■■■■
Methodology & Equations	4.0	Good	■■■■■
Statistical Analysis	4.5	Excellent	■■■■■
Results & Evidence	4.2	Good	■■■■■
Discussion & Conclusions	3.2	Acceptable	■■■■■
Figures & Tables	4.0	Good	■■■■■
Scientific Writing	3.2	Acceptable	■■■■■
References & Citations	3.2	Acceptable	■■■■■

Ethics & Transparency	1.5	Very Weak	■■■■■
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Detailed Agent Reviews

TitleAbstractKeywordsReviewer

Title, Abstract, Keywords

Score: 5.0 / 5.0 | Confidence: 100%

Strengths:

- [TITLE] The title is specific, informative, and includes key terms for discoverability. It clearly reflects the contribution of the study.
- [ABSTRACT] The abstract follows a structured format with clear background, objective, methods, results, and conclusions. It quantifies results and avoids vague language.
- [KEYWORDS] The keywords are comprehensive, include domain-specific terms, and avoid duplication of title words.
- [TITLE] The title clearly signals the core method and application domain.
- [ABSTRACT] The abstract provides substantial methodological and validation detail.

StructureReviewer

Manuscript structure and logical coherence

Score: 2.5 / 5.0 | Confidence: 95%

Strengths:

- Clear title and keywords
- Well-defined methodology section
- Experimental results presented with figures

Weaknesses:

- Missing literature review section
- Inconsistent section labeling (e.g., 'section_vi' and 'section_vii' are repeated content)
- Research problem and hypothesis not clearly stated in the introduction
- Objectives not explicitly stated or aligned with conclusions
- Poorly structured flow between sections
- Redundant content in 'section_vi' and 'section_vii'

Major Comments:

- The manuscript is missing a critical 'literature review' section, which is essential for establishing the context and significance of the research. This omission severely undermines the logical flow and coherence of the manuscript.
- Sections 'section_vi' and 'section_vii' contain identical content, which indicates a significant duplication error. This redundancy compromises the manuscript's integrity and should be resolved.
- The research problem and hypothesis are not clearly stated in the introduction. A Q1 journal requires a clear and explicit statement of the research problem and hypothesis to justify the study.
- The objectives of the study are not explicitly stated in the introduction, nor are they clearly addressed in the conclusions. This disconnect between objectives and conclusions weakens the logical coherence of the manuscript.
- The flow between sections is inconsistent and lacks proper transitions. The manuscript jumps between sections without clear logical progression, which is not acceptable for a Q1 journal.

Minor Comments:

- The title and keywords are well-defined, which is a positive aspect for indexing and searchability.
- The methodology section is well-structured, but it lacks sufficient detail to ensure reproducibility.
- The experimental results are presented with figures, but the quantitative claims are not fully supported by the data or statistical analysis.
- The abstract is missing, which is a critical component for a scientific manuscript.

Recommendations:

- Add a comprehensive 'literature review' section to provide context and justify the research problem.
- Remove or revise the duplicated content in 'section_vi' and 'section_vii' to eliminate redundancy.
- Clearly state the research problem and hypothesis in the introduction to align with the objectives and conclusions.
- Explicitly define the objectives in the introduction and ensure they are fully addressed in the conclusions.
- Improve the logical flow between sections by adding clear transitions and ensuring each section logically leads to the next.
- Include a detailed abstract to summarize the key contributions and findings of the study.

MethodologyReviewer

Methodology, experimental design, equations — holistic review

Score: 4.0 / 5.0 | Confidence: 95%

Strengths:

- The paper presents a novel physics-informed EANN framework with a clear multi-objective formulation.
- The experimental setup is well-documented, including the five-story building prototype and the use of accelerometers.
- The statistical validation methods (Kruskal-Wallis, Wilcoxon rank-sum tests, nonparametric bootstrap) are appropriate for the nonparametric nature of the data.
- The discussion effectively links the results to practical implications for structural health monitoring and control design.

Weaknesses:

- The 'Experiments' section lacks a clear description of the exact implementation of the EANN framework, including the neural network architecture and training procedure.
- The 'Methodology' section does not fully explain the integration of physical constraints into the neural network training process.
- The paper does not provide a detailed description of the evolutionary algorithm implementation, such as the selection, crossover, and mutation operators used.
- The 'Discussion' section could have more explicitly addressed the limitations of the proposed method and its potential for real-world application.

Major Comments:

- The paper should clarify the exact implementation of the EANN framework, including the neural network architecture and training procedure, to ensure reproducibility.
- The integration of physical constraints into the neural network training process should be more clearly explained to demonstrate how the framework enforces physical plausibility.
- The evolutionary algorithm implementation details (e.g., selection, crossover, mutation) should be provided to allow for replication of the results.

Minor Comments:

- Equation (3) should be fully defined with the expressions for matrices A and B to ensure clarity.
- The 'Experiments' section could benefit from a more detailed description of the data preprocessing steps and their justification.
- The 'Discussion' section could include a more thorough comparison of the EANN framework with existing methods in terms of computational efficiency and scalability.

Recommendations:

- Include a detailed description of the neural network architecture, including the number of layers, neurons, activation functions, and training hyperparameters.
- Provide a more comprehensive explanation of the evolutionary algorithm implementation, including the selection, crossover, and mutation operators.

- Clarify the exact integration of physical constraints into the neural network training process, including how the constraints are enforced and their impact on the optimization.

StatisticsReviewer

Statistical methods, tests, and reporting quality

Score: **4.5 / 5.0** | Confidence: 95%

Strengths:

- The use of Kruskal-Wallis and Wilcoxon rank-sum tests with Holm correction is appropriate for non-parametric comparisons of multiple groups.
- Bootstrap confidence intervals are reported, which is a strength for non-normal data.
- The coefficient of variation (CV) is reported and interpreted correctly, providing insight into the practical significance of parameter variability.
- RMSE, MAE, and R^2 are used and interpreted appropriately for evaluating model accuracy.
- Pareto front analysis and hypervolume indicators are used to assess multi-objective trade-offs, which is appropriate for the context.
- The manuscript reports 30 independent runs, which is a reasonable number for assessing robustness in stochastic optimization.

Weaknesses:

- The manuscript does not report p-values for the Wilcoxon rank-sum post-hoc tests, which are essential for determining statistical significance.
- The manuscript does not report the number of bootstrap resamples used to compute confidence intervals, which is important for assessing the reliability of the bootstrap method.
- The manuscript does not report the exact values of the Kruskal-Wallis statistic or the corresponding p-values, which are necessary for full transparency.
- The manuscript does not report the exact values of the coefficient of variation (CV) for each parameter, which would allow for more detailed interpretation.
- The manuscript does not report the exact values of the hypervolume and spread indicators, which are critical for assessing the quality of the multi-objective optimization.

Major Comments:

- The manuscript should report p-values for the Wilcoxon rank-sum post-hoc tests to fully support the statistical claims made in the results section.
- The manuscript should report the exact values of the Kruskal-Wallis statistic and the corresponding p-values to ensure transparency and reproducibility.
- The manuscript should report the number of bootstrap resamples used to compute confidence intervals, as this is essential for assessing the validity of the bootstrap method.

Minor Comments:

- The manuscript could benefit from reporting the exact values of the coefficient of variation (CV) for each parameter to allow for more detailed interpretation.
- The manuscript could benefit from reporting the exact values of the hypervolume and spread indicators to allow for more detailed interpretation of the multi-objective optimization results.

Recommendations:

- Report p-values for the Wilcoxon rank-sum post-hoc tests to fully support the statistical claims made in the results section.
- Report the exact values of the Kruskal-Wallis statistic and the corresponding p-values to ensure transparency and reproducibility.
- Report the number of bootstrap resamples used to compute confidence intervals, as this is essential for assessing the validity of the bootstrap method.

- Report the exact values of the coefficient of variation (CV) for each parameter to allow for more detailed interpretation.
- Report the exact values of the hypervolume and spread indicators to allow for more detailed interpretation of the multi-objective optimization results.

FiguresTablesEquationsReviewer

Figures, tables, equations — quality and coherence with text

Score: 4.0 / 5.0 | Confidence: 62%

Strengths:

- 6 figure references were detected in the manuscript.
- 10 table references were detected in the manuscript.
- 52 sequentially numbered equations were detected, suggesting a substantial mathematical formulation.

Weaknesses:

- The model response could not be used reliably, so this fallback verifies presence and numbering but cannot judge visual quality or table body completeness.

Minor Comments:

- The model response could not be used reliably, so this fallback verifies presence and numbering but cannot judge visual quality or table body completeness.

Recommendations:

- Manually inspect captions, axes, units, table headers, and variable definitions in the final PDF.

ResultsReviewer

Results validity and empirical support

Score: 4.2 / 5.0 | Confidence: 85%

Strengths:

- The results clearly address the stated objectives of the study, including the performance of the EANN framework in comparison to GA and PSO baselines.
- Quantitative results are well-supported with MSE, RMSE, CV, and hypervolume metrics, which are relevant to the multi-objective optimization context.
- The manuscript includes a thorough statistical validation using Kruskal-Wallis, Wilcoxon rank-sum tests, and nonparametric bootstrap confidence intervals.
- The comparison with state-of-the-art methods (GA and PSO) is explicitly detailed, including the use of 30 independent runs for robustness.
- Negative results, such as the reduced parametric dispersion and denser Pareto front, are honestly reported and tied to the control-oriented implications.

Weaknesses:

- The manuscript lacks specific details on the exact values of the MSE, RMSE, and CV metrics for each floor and method, which are only referenced in the abstract and not fully detailed in the results section.
- The Pareto front analysis and hypervolume/spread indicators are mentioned but not fully elaborated with visual representation or detailed quantitative comparison with GA and PSO.
- The exact values of the nonparametric bootstrap confidence intervals are not provided, which weakens the statistical validation claims.
- The methodology for the KNEESELECT function and the local refinement step (lines 23-25) is not clearly explained in the results section, making it difficult to assess the validity of the compromise solution selection.
- The specific values of the baseline comparisons (GA and PSO) are not fully detailed in the results section, which limits the ability to assess the significance of the EANN's performance improvements.

Major Comments:

- The results section should include detailed quantitative metrics (e.g., MSE, RMSE, CV) for each floor and method, as these are critical for assessing the performance of the EANN framework.
- The Pareto front and hypervolume/spread indicators should be presented with visual representation and detailed quantitative comparison with GA and PSO to support the claims of a denser and more uniformly distributed front.
- The statistical validation section should include the exact values of the nonparametric bootstrap confidence intervals to strengthen the empirical support for the results.

Minor Comments:

- The results section could benefit from a more structured presentation of the quantitative results, such as a table summarizing the performance metrics for each method and floor.
- The methodology for the KNEESELECT function and the local refinement step should be explained in more detail in the results section to ensure reproducibility and clarity.
- The exact values of the baseline comparisons (GA and PSO) should be included in the results section to provide a clearer picture of the EANN's performance relative to the baselines.

Recommendations:

- Include a table with detailed performance metrics (MSE, RMSE, CV, etc.) for each floor and method to enhance clarity and reproducibility.
- Add visual representation of the Pareto front and hypervolume/spread indicators to support the claims of improved multi-objective performance.
- Provide the exact values of the nonparametric bootstrap confidence intervals to strengthen the statistical validation of the results.
- Clarify the methodology for the KNEESELECT function and the local refinement step in the results section to ensure transparency and reproducibility.

DiscussionConclusionsReviewer

Discussion depth and conclusions validity

Score: **3.2 / 5.0** | Confidence: 90%

Strengths:

- The Discussion section effectively contextualizes the results within the broader scope of structural identification methods, particularly highlighting the advantages of EANN over GA and PSO.
- The Conclusions section clearly states the contribution of the proposed framework and its potential impact on structural health monitoring and adaptive control systems.
- The manuscript provides a clear and specific statement of the contribution to the field, which is essential for Q1 journal acceptance.

Weaknesses:

- The Discussion section lacks a comprehensive comparison with prior work, focusing only on citation without detailed analysis of how the current findings align or diverge from existing literature.
- The Discussion does not adequately address unexpected findings or anomalies in the results, such as the occasional lower median error values achieved by GA, which could suggest limitations in the EANN approach.
- The Discussion does not clearly state the limitations of the study, such as the computational cost of EANN or the assumption of identical conditions across all experimental runs.
- The Conclusions section overgeneralizes by stating that EANN 'enables near real-time integration into structural health monitoring pipelines' without quantifying or justifying this claim with specific performance metrics.
- The Conclusions section does not specify concrete future work directions, such as exploring hybrid models, extending to more complex structures, or validating under different excitation conditions.

Major Comments:

- The Discussion section fails to engage with the broader literature on structural identification methods, which is critical for establishing the novelty and significance of the proposed EANN framework.
- The manuscript does not address the potential limitations of the EANN approach, such as computational cost, which is a key factor in real-world deployment.

- The overgeneralization in the Conclusions section undermines the credibility of the claims made about the practical utility of the EANN framework.

Minor Comments:

- The Discussion section could benefit from a more detailed comparison with prior work, including specific references to how the current findings align or differ from existing studies.
- The Conclusions section could be more specific in its statements about the practical implications of the EANN framework.

Recommendations:

- Include a detailed comparison with prior work in the Discussion section, highlighting how the EANN framework advances or differs from existing methods.
- Clarify the limitations of the study, such as computational cost and assumptions about experimental conditions, to provide a balanced view of the proposed approach.
- Avoid overgeneralization in the Conclusions section by grounding claims in specific performance metrics and avoiding vague statements about real-time integration.
- Specify concrete future work directions, such as exploring hybrid models, extending to more complex structures, or validating under different excitation conditions.

WritingReviewer

Scientific writing quality, clarity, and style

Score: **3.2 / 5.0** | Confidence: 85%

Strengths:

- Clear and structured presentation of methodology
- Robust statistical validation with nonparametric tests
- Well-defined experimental setup and benchmarking
- Clear connection between results and control-oriented applications

Weaknesses:

- Overuse of passive voice
- Unexplained acronyms
- Vague quantifiers
- Lack of technical term definitions
- Run-on sentences
- Inconsistent terminology
- Weak paragraph structure

Major Comments:

- The manuscript contains several instances of passive voice that could be rephrased for clarity and directness.
- Acronyms such as EANN, PSO, and GA are used without explanation, which may confuse readers unfamiliar with the field.
- The phrase 'significant' is used without numerical support, which weakens the claim.
- The sentence 'This balance between model simplicity and descriptive accuracy makes lumped-parameter models particularly attractive for control applications' is a run-on and lacks clear structure.
- The paragraph 'The proposed EANN framework was evaluated along four dimensions — accuracy and statistical inference, multi-objective quality, parametric robustness, and computational deployment — each of which carries direct implications for control-oriented structural modeling' is overly long and lacks clear topic sentences.

Minor Comments:

- The phrase 'This paper proposes a novel...' is overused and should be avoided.
- The sentence 'The EANN framework is rigorously compared against a multi-objective Genetic Algorithm (GA) and a Particle Swarm Optimization (PSO) baseline over 30 independent stochastic runs per method' is a run-on and

could be split for clarity.

- The sentence 'From a control-oriented perspective, reduced parametric variability translates directly into tighter eigenvalue uncertainty bounds on the identified state matrix, enhancing reliability for structural health monitoring, observer design, vibration control, and seismic engineering applications' is a run-on and lacks clear transitions.
- The sentence 'The nonparametric tests reported in Section V-B confirm that performance differences among GA, PSO, and EANN are statistically significant across all metrics ($p < 0.001$)' is a run-on and could be split for clarity.

Recommendations:

- Define acronyms such as EANN, PSO, and GA on their first use.
- Avoid the overuse of passive voice and rephrase sentences for active voice where appropriate.
- Replace vague quantifiers like 'significant', 'many', and 'several' with specific numerical values.
- Split long sentences into shorter, more digestible ones with clear topic sentences and transitions.
- Ensure that all technical terms are defined on their first use.
- Maintain consistent terminology throughout the manuscript.
- Use connectors and transitions effectively to improve the logical flow between paragraphs and sections.

ReferencesReviewer

References quality, recency, and relevance

Score: **3.2 / 5.0** | Confidence: 80%

Strengths:

- The references are consistently formatted in IEEE style.
- A mix of journal articles and conference proceedings is included, which is appropriate for the field.
- Several recent references (2021-2023) are included, which is good for a fast-moving field like structural health monitoring and control.

Weaknesses:

- The references section lacks seminal foundational works from the 1990s and early 2000s, which are critical for establishing the context of lumped mass-spring-damper models and their use in structural health monitoring.
- The references section does not include key foundational works on seismic design regulations, such as those from Mexico City, which are mentioned in the introduction.
- The references section lacks a comprehensive review of the state-of-the-art in structural parameter identification, which is a key topic in the introduction.

Major Comments:

- The absence of seminal foundational works from the 1990s and early 2000s undermines the literature review and the contextualization of the current work. These works are essential for establishing the background and significance of the study.
- The references section does not adequately address the historical development of lumped mass-spring-damper models and their application in structural health monitoring and seismic design regulations.
- The lack of a comprehensive review of state-of-the-art methods in structural parameter identification weakens the manuscript's contribution to the field.

Minor Comments:

- The citation of [29] Chopra's book is not directly referenced in the text, which may suggest a lack of direct engagement with the source.
- The reference [34] is cut off and not fully visible, which limits the ability to assess its relevance and contribution.

Recommendations:

- Include seminal foundational works from the 1990s and early 2000s, such as the original papers on lumped mass-spring-damper models and their use in structural health monitoring.
- Add references to key foundational works on seismic design regulations, such as those from Mexico City, which are mentioned in the introduction.

- Include a comprehensive review of state-of-the-art methods in structural parameter identification, including recent and foundational works.
- Ensure that all cited references are directly referenced in the text and that the citations are relevant to the claims made.

EthicsLimitationsReviewer

Research ethics, limitations, and transparency

Score: **1.5 / 5.0** | Confidence: 80%

Strengths:

- The methodology is well-structured and presents a novel approach to structural parameter identification using a physics-informed EANN framework.
- The experimental results are supported by quantitative metrics (RMSE, MAE, R2) and visualizations (boxplots, scatter plots), which enhance the credibility of the findings.

Weaknesses:

- The manuscript lacks comprehensive reporting on ethical considerations, which is critical for ensuring the research adheres to COPE and IEEE standards.
- The limitations section is either missing or not adequately addressed, which undermines the validity of the conclusions and the generalizability of the findings.
- The reproducibility of the results is not addressed, as no data or code availability statement is provided, which is a fundamental requirement for scientific transparency.

Major Comments:

- The manuscript fails to meet the ethical reporting standards required by Q1 journals. Ethical approval, informed consent, data availability, conflict of interest, and funding disclosure are all missing, which is a major issue for publication.
- The limitations section is either absent or not properly addressed. The manuscript does not explicitly state the limitations, nor does it assess their impact on the conclusions, which is a critical omission for a high-impact journal.

Minor Comments:

- The methodology is well-explained, but the lack of detailed information on the experimental setup (e.g., exact dimensions, sensor specifications, and environmental conditions) could affect the reproducibility of the results.
- The discussion section could benefit from a more detailed analysis of the implications of the findings and how they align with existing literature.

Recommendations:

- Include a statement on ethical approval, informed consent, data availability, conflict of interest, and funding sources.
- Provide a detailed limitations section that explicitly states the limitations, their impact on the conclusions, and how they affect the generalizability of the findings.
- Include a reproducibility statement that specifies the availability of data and code, which is essential for transparency and validation.

Consolidated Major Comments

These issues must be addressed before the manuscript can be accepted.

- [1] The manuscript is missing a critical 'literature review' section, which is essential for establishing the context and significance of the research. This omission severely undermines the logical flow and coherence of the manuscript.
- [2] Sections 'section_vi' and 'section_vii' contain identical content, which indicates a significant duplication error. This redundancy compromises the manuscript's integrity and should be resolved.
- [3] The research problem and hypothesis are not clearly stated in the introduction. A Q1 journal requires a clear and explicit statement of the research problem and hypothesis to justify the study.
- [4] The objectives of the study are not explicitly stated in the introduction, nor are they clearly addressed in the conclusions. This disconnect between objectives and conclusions weakens the logical coherence of the manuscript.
- [5] The flow between sections is inconsistent and lacks proper transitions. The manuscript jumps between sections without clear logical progression, which is not acceptable for a Q1 journal.
- [6] The paper should clarify the exact implementation of the EANN framework, including the neural network architecture and training procedure, to ensure reproducibility.
- [7] The integration of physical constraints into the neural network training process should be more clearly explained to demonstrate how the framework enforces physical plausibility.
- [8] The evolutionary algorithm implementation details (e.g., selection, crossover, mutation) should be provided to allow for replication of the results.
- [9] The manuscript should report p-values for the Wilcoxon rank-sum post-hoc tests to fully support the statistical claims made in the results section.
- [10] The manuscript should report the exact values of the Kruskal-Wallis statistic and the corresponding p-values to ensure transparency and reproducibility.
- [11] The manuscript should report the number of bootstrap resamples used to compute confidence intervals, as this is essential for assessing the validity of the bootstrap method.
- [12] The results section should include detailed quantitative metrics (e.g., MSE, RMSE, CV) for each floor and method, as these are critical for assessing the performance of the EANN framework.

Consolidated Minor Comments

These are suggestions that would improve the manuscript quality.

- [1] The title and keywords are well-defined, which is a positive aspect for indexing and searchability.
- [2] The methodology section is well-structured, but it lacks sufficient detail to ensure reproducibility.
- [3] The experimental results are presented with figures, but the quantitative claims are not fully supported by the data or statistical analysis.
- [4] The abstract is missing, which is a critical component for a scientific manuscript.
- [5] Equation (3) should be fully defined with the expressions for matrices A and B to ensure clarity.
- [6] The 'Experiments' section could benefit from a more detailed description of the data preprocessing steps and their justification.
- [7] The 'Discussion' section could include a more thorough comparison of the EANN framework with existing methods in terms of computational efficiency and scalability.
- [8] The manuscript could benefit from reporting the exact values of the coefficient of variation (CV) for each parameter to allow for more detailed interpretation.
- [9] The manuscript could benefit from reporting the exact values of the hypervolume and spread indicators to allow for more detailed interpretation of the multi-objective optimization results.
- [10] The model response could not be used reliably, so this fallback verifies presence and numbering but cannot judge visual quality or table body completeness.

[11] The results section could benefit from a more structured presentation of the quantitative results, such as a table summarizing the performance metrics for each method and floor.

[12] The methodology for the KNEESELECT function and the local refinement step should be explained in more detail in the results section to ensure reproducibility and clarity.

Specific Improvement Recommendations

1. Add a comprehensive 'literature review' section to provide context and justify the research problem.
2. Remove or revise the duplicated content in 'section_vi' and 'section_vii' to eliminate redundancy.
3. Clearly state the research problem and hypothesis in the introduction to align with the objectives and conclusions.
4. Explicitly define the objectives in the introduction and ensure they are fully addressed in the conclusions.
5. Improve the logical flow between sections by adding clear transitions and ensuring each section logically leads to the next.
6. Include a detailed abstract to summarize the key contributions and findings of the study.
7. Include a detailed description of the neural network architecture, including the number of layers, neurons, activation functions, and training hyperparameters.
8. Provide a more comprehensive explanation of the evolutionary algorithm implementation, including the selection, crossover, and mutation operators.
9. Clarify the exact integration of physical constraints into the neural network training process, including how the constraints are enforced and their impact on the optimization.
10. Report p-values for the Wilcoxon rank-sum post-hoc tests to fully support the statistical claims made in the results section.
11. Report the exact values of the Kruskal-Wallis statistic and the corresponding p-values to ensure transparency and reproducibility.
12. Report the number of bootstrap resamples used to compute confidence intervals, as this is essential for assessing the validity of the bootstrap method.